AFC–7780

Sub. Code
1MCH4E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fourth Semester

Chemistry – Elective

CORROSION SCIENCE

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are classification of corrosion?

2. Define the term erosion corrosion.

3. How is differential aeration corrosion prevented?

4. What is atmospheric corrosion?

5. Write different type of electrode.

6. Define the term EMF.

7. What is water activated battery?

8. Write electrochemical series of lithium cells.

9. Give the various types of fuel cell.

10. What is oxygen reduction?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Write short note on passivity of metals.

Or

(b) Discuss about the corrosion control methods.

12. (a) Describe the various factors affect atmospheric corrosion.

Or

(b) What is biological corrosion? Give the mechanism and control methods.

13. (a) Describe the applications of primary and secondary cells.

Or

(b) Discuss in detail non – aqueous and molten salt electrolytes.

14. (a) Write and short note on lead – acid battery.

Or

(b) Explain lithium ion battery.

15. (a) Describe the electro catalysis of hydrogen oxidation.

Or

(b) Write short note mercury porosimetry and limiting current density.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the following corrosions :

    (a) Stress

    (b) Crevice

    (c) Fretting

    (d) Film form corrosion.

17. Explain the important factors which influence the rate of corrosion metal.

18. (a) Discuss basic principle and criteria for a selection of anodes and cathodes.

    (b) Explain thermodynamic reversibility.

19. Describe in detail water activated battery like (a) Ni – Cd (b) Ni - Fe (c) Ni – metal hydride.

20. Write note on the following fuel cell :

    (a) Solid oxide

    (b) Alkaline

    (c) Phosphoric fuel cell system.

AFC–7765

Sub. Code
1MCH1C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Chemistry

INORGANIC CHEMISTRY – I

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Distinguish between sigman and Π bonds.

2. Define bond order.

3. Electron affinity of fluorine is less than that of chlorine. Explain.

4. What is levelling affect?

5. What are n-type and p-type semiconductors?

6. Define colour centers in solids.

7. What are isopoly acids? Give an example.

8. What are cyclic silicates? Give an example.

9. Mention any two ores of Beryllium.

10. Calculate the EAN of Ni in $Ni(CO)_4$.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Construct wave functions of Sp^3 hybrid orbitals.

Or

(b) Using MO theory, calculate bond order of NO and NO^+.

12. (a) Discuss in detail about the effect of steric factor on the strength of acids and bases.

Or

(b) Explain the following, giving reasons :

(i) Bond energy of NO^+ in higher than of NO.

(ii) Electron affinity of noble gases in zero.

13. (a) Discuss briefly the point defect in crystal.

Or

(b) Write a note on high temperature super conductor.

14. (a) Discuss the structure of $[MO_8O_{26}]^{4-}$ ion.

Or

(b) Write a note on Wade's rule as applied to carboranes and how are they classified?

15. (a) Discuss the extraction of thorium from its ore.

Or

(b) Describe various reactions of ferrocene which establish its aromatic character.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Differentiate molecular orbitals from atomic orbitals.

(b) Discuss the Mo diagram and molecular orbital treatment of carbonate ion $\left(CO_3^{2-}\right)$. (4+6)

17. Explain the following :

(a) Dimethylamine is more basic than trimethylamine.

(b) $HCLO_4$ is more acidic than $HCLO_3$.

(c) K^+ ion is smaller than Cl^- eventhough both are isoelectronic.

(d) Bond length of N_2 is smaller than O_2. (4 × 2 ½ = 10)

18. (a) Describe the role of semiconductor in solar energy conversion.

(b) Discuss the free electron theory of solids.

19. (a) What are silicones? Discuss their structure and applications.

(b) Discuss the structure of borazine.

20. (a) Explain the structure and bonding in metal carbonyl.

(b) Describe the isolation, properties and uses of beryllium.

AFC–7766

Sub. Code
1MCH1C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Chemistry

PHYSICAL CHEMISTY – I

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is Wein Effect.

2. Define activity coefficients.

3. What is hydrogen over voltage?

4. Define Half wave potentials.

5. What do you understand by partial molar quantities?

6. Define chemical potential.

7. What is an ensemble as used in statistical mechanics?

8. What is population inversion?

9. Define reaction isotherm.

10. State Gibb's phase rule.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Write short notes on:

(i) Interionic attraction

(ii) Thickness of ionic atmosphere.

Or

(b) Explain the following:

(i) Debye-Falkenhagen effect

(ii) Meanionic Activity

12. (a) Explain the Perrin and Stern models of electrical double layer.

Or

(b) Derive Butlervolmer equation.

13. (a) Derive Maxwell relations as given below:

(i) $\left[\frac{\partial T}{\partial V}\right]_S = -\left[\frac{\partial P}{\partial S}\right]_V$

(ii) $\left[\frac{\partial S}{\partial P}\right]_T = -\left[\frac{\partial V}{\partial T}\right]_P$

Or

(b) Derive Gibbs-Duhemequation.

14. (a) Write a note on thermodynamic properties from partition functions.

Or

(b) Derive an expression for the Bose-Einstein statistics.

15. (a) Give an account of the calculation of reaction free energy from electrochemical data.

Or

(b) Derive thermodynamically phase rule equation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Derive Debye–Huckel-onsager conductance equation and its experimental verification.

17. (a) Write a note on cyclic voltammetry.

(b) Give any two applications of polarography.

18. (a) Explain the concept of entropy.

(b) How will you determine the fugacity of a real gases?

19. (a) Derive Boltzmann –Planck equation.

(b) Derive an expression for the Fermi-Dirac Statistics.

20. (a) At 25°C, the $\Delta G°$ for the reaction. $H_2(g) + I_2(g) \rightarrow 2HI(g)$, is –3.1 Kcal. Calculate ΔG when hydrogen gas at 0.1 atm and iodine vapour at 0.1 atm react to form hydrogeniodide at 10 atm, and 25°C.

(b) What is incongruent melting point? Explain with suitable example.

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AFC–7767

Sub. Code
1MCH1E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Chemistry – Elective

MEDICINAL AND PHARMACEUTICAL CHEMISTRY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Explain drug elimination.

2. Give any two factors affecting the absorption of drugs.

3. Draw the structure of cephalosphorin.

4. What are the adverse effects of chloramphenicol.

5. Write the chemical names of:
   (a) Chlorambucil
   (b) Methotrexate.

6. What are antitubercular drugs? Give examples.

7. Draw the structure and give the adverse effects of hydralazine.

8. Write the chemical formula and uses of sodium nitroprusside.

9. Give any three applications of UV-visible spectroscopy.

10. Explain the principles of TLC.

Part B (5 × 5 = 25)

Answer **all** the questions, choosing either (a) or (b).

11. (a) Write notes on:

(i) Drug distribution

(ii) Absorption of drugs.

Or

(b) Write in detail about lead compounds.

12. (a) Draw the structure and discuss the SAR of pencillin.

Or

(b) Draw the structure and explain the structural features of erythromycin.

13. (a) Give the synthesis and assay of cisplatin.

Or

(b) Write a note on the following:

(i) Primaquine

(ii) Progunail pyrimethamine.

14. (a) Draw the structure and discuss the therapeutic action of nifedipine.

Or

(b) Give the synthesis and discuss the SAR of clonidine.

15. (a) Explain the principle of mass spectroscopy and give its applications.

Or

(b) Write a short on high performance liquid chromatography.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Explain in detail about drug metabolism.

(b) Discuss in detail about the steps involved in drug designing.

17. (a) Draw the structure and give the structure activity relationship of streptomycin.

(b) Give the structural features and therapeutic uses of terramycin.

18. Write a note on the synthesis and assay of the following chemotherapeutic agents

(a) Ifosfamide

(b) Chloroquine.

19. (a) Write about the SAR of captopril.

(b) Give the synthesis and therapeutic action of sodium nitroprusside.

20. Explain the principle, instrumentation and applications of IR spectroscopy.

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AFC–7769

Sub. Code
1MCH2C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Chemistry

INORGANIC CHEMISTRY — II

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What are ambidentate ligands? Give examples.

2. Is the complex $[Co(NH_3)_6]Br_3$ paramagnetic or diamagnetic?

3. What are labile complexes?

4. Define complementary electron transfer reaction.

5. Define Bohr effect.

6. What is ferridoxin? Give its function.

7. List the role of Zn and Co in biological processes.

8. What is meant by inhibition?

9. What is oxo-process? Give its importance.

10. Define oxidative addition reaction.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain how the stability constant of a complex can be determined by spectrophotometric method.

Or

(b) Draw the d – orbital splitting diagram, show the electron occupancies, calculate the CFSE and predict the magnetic moment in BM for (i) $[Fe(CN)_6]^{4-}$ and (ii) $[Mn(H_2O)_6]^{2+}$. (3 + 2)

12. (a) Explain the Π – bonding theory of trans effect.

Or

(b) Discuss the mechanism of inner sphere electron transfer reaction with an example pointing out the evidences and consequences.

13. (a) Discuss the structure and functions of non-sulphur proteins.

Or

(b) Write short notes on photosynthetic applications of chlorophyll.

14. (a) Discuss the structure and function of carboxy peptidase.

Or

(b) State and explain the active centre present in the MO-enzyme involved in nitrogen fixation.

15. (a) What is Wacker process? Discuss its mechanistic aspects.

Or

(b) Explain the role of $HCo(Co)_4$ in the hydroformylation process.

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. (a) Draw the MO energy level diagram for an octahedral complex and label the various orbitals.

    (b) State Jahn-Teller theorem. Apply the theorem to d^9 system. (5 + 5)

17. (a) Explain the mechanism of acid hydrolysis reaction.

    (b) Discuss how the coordination compounds and synthesised using electron transfer reaction. (5 + 5)

18. (a) Explain the functioning of hemoglobin and myoglobin with respect to the following :

    (i) High spin-low spin trigger (3)

    (ii) Cooperativity effect. (3)

    (b) Write short notes on synthetic oxygen carriers. (4)

19. (a) Discuss the structure of vitamin B_{12} and explain the non-enzymatic chemistry of B_{12} coenzyme.

    (b) Explain the function of carbonic anhydrase. (5 + 5)

20. What is Ziegler-Natta catalyst? How does it catalyse the polymerisation of olefins? What is its importance?

AFC–7770

Sub. Code
1MCH2C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Chemistry

PHYSICAL CHEMISTRY – II

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. Calculate the energy in kJ mol^{-1} of photon of wave length 600 nm.

2. Draw the molecular energy level diagram showing rotational, vibrational and electronic transitions.

3. What are overtones and combination bands?

4. Electronic excitations give rise to broad bands rather than sharp lines. Account for it.

5. When does a hyperfine structure arise?

6. What is meant by nuclear quadrupolar interaction?

7. Write the symmetry elements and assign point group for the molecules XeF_4 and benzene.

8. What are symmetry elements and symmetry operations?

9. Write the secular determinant for benzene.

10. How many normal modes of vibrations would you expect for CO_2 and benzene molecules?

Section B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the effect of isotopic substitution in rotational spectrum of CO.

Or

(b) In the pure rotational spectrum of HCl, the lines are separated by 20.80 cm^{-1}. What is the internuclear distance in the molecule?

12. (a) Write notes on Laser Raman spectra.

Or

(b) Explain the applications of IR and Raman spectra in the structural determination of planar, pyramidal and Non-symmetric AB_3 type molecules.

13. (a) Explain the basic principles of ESR spectroscopy.

Or

(b) Write the energy level expressions and calculate NQR frequency ratios for ^{35}Cl and ^{27}Al isotopes.

14. (a) Construct the character table for C_{2h} point group.

Or

(b) Show that all the representations of C_{3v} group are mutually orthogonal.

15. (a) Determine the number of normal modes of vibrations of BF_3 molecule and identify IR and Raman active modes.

Or

(b) State the group theoretical selection rules and write the transition integrals for IR and Raman spectra.

Section C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Discus the information obtained from rotations spectra.

17. (a) Discuss the polarizability theory of the Raman effect.

(b) What are P and R branch lines? Show them in a rotational vibrational diagram. (5 + 5)

18. (a) Write notes on FT-IR spectroscopy.

(b) What is zero field splitting in ESR? Explain. (5 + 5)

19. (a) State the Great Orthogonality theorem.

(b) What are the important rules of irreducible representations? (5 + 5)

20. Obtain the four energy levels and draw the HMO diagram with appropriate designation of irreducible representations for 1, 3 butadiene molecule.

AFC–7771

Sub. Code
1MCH2E2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Chemistry

Elective – INDUSTRIAL CHEMISTRY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is unit operation?

2. Write a note on designing of chemical plant.

3. What is cracking? Illustrate with an example.

4. What are petro chemicals? Give example.

5. Mention the name of cement industries in India.

6. State the raw materials and formulations of detergent.

7. List out the essential raw materials used in glass making.

8. Distinguish between drying oil and semi drying oils.

9. How is B.H.C prepared? Mention its uses.

10. In what basis, the pesticides are classified?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) What are the basic requirements of industrial reactors?

Or

(b) Explain the principle of homogeneous and heterogeneous process.

12. (a) Describe the characteristics of raw materials and their resources.

Or

(b) Explain the classification of fuels with example.

13. (a) (i) What are the raw materials used in the manufacture of economics?

(ii) What are the types of pottery products?

Or

(b) Write notes on :

(i) Selting of cement

(ii) Neutron absorbing glass.

14. (a) Describe the formulation and manufacture of fire-works.

Or

(b) Write in detail the electroplating method and glavanising.

15. (a) How is cane sugar manufactured?

Or

(b) What are the classifications of fungicides? Explain with examples.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Write a note on the modeling of chemical plant.

    (b) Explain in detail the choice and selectivity of reactor.

17. (a) Describe the methods of material concentrations.

    (b) Explain the composition and applications of coal gas, oil gas and producer gas.

18. (a) Discuss the various stages involved in the manufacturing of ceramics.

    (b) How are the following manufactured?

    (i) Triple superphosphate

    (ii) Colored glasses.

19. (a) Explain briefly the electro-thermal and electro-chemical processes with example.

    (b) Discuss the manufacture of safety matches.

20. (a) Describe insecticicles with suitable example.

    (b) How is super recovered from molasses? Mention sugar industries in India.

AFC–7772

Sub. Code
1MCH3C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015.

Third Semester

Chemistry

ORGANIC CHEMISTRY — III

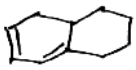
(CBCS – 2011 onwards)

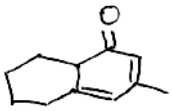
Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

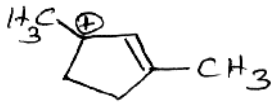
1. Calculate the λ_{max} for the following compounds. (1+1)

   (a) 

   (b) 

2. Which spectroscopic technique you would use to distinguish between $CH_3(CH_2)_{10}COOE_t$ and $CH_3(CH_2)_{11}COOE_t$? Explain your answer.

3. The ^{1}HNMR spectrum of the following cation shows only three signals with area ratio 6:4:1. Explain.



4. How will you proceed to distinguish between samples of para-xylene and meta-xylene by ^{13}C NMR spectroscopy?

5. What are the differences between packed column chromatography and capillary column chromatography?

6. What do you understand by circular birefringence?

7. How is dehyroepiandrosterone converted in to androsterone?

8. What happens when cholesterol is distilled with selenium?

9. Give reason for the stability of triphenyl methyl radical.

10. Comment on Gomberg – Bachmann reaction.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Write brief note on ;

(i) Metastable ion.

(ii) Retro Diels - Alder reaction. (2.5+2.5)

Or

(b) (i) In the infrared spectrum of the methyl ester of 0 - chlorobenzonic acid two bands for the carbonyl stretching appear at 1685 and $1708\,cm^{-1}$. Retionalise this observation.

(ii) Give the fragmentation pattern of *n* - batylmethy Ketone. (3+2)

12. (a) (i) What do you understand by the term splitting of the signals? Explain with an example.

(ii) ^{13}C carbonyl chemical shift of acetophenone is 196 ppm, where as that of 2, 6-dimethylacetophenone is 206 ppm. Explain. (3+2)

Or

(b) Give the structural formula for the substance having the following 'H NMR data.

Triplet, area 3, at δ 1.03 ; multiplet of at least six lines, area 2, at δ 2.07 ; triplet are 2, at δ 4.38. Molecular formula $C_3H_7NO_2$.

13. (a) How are the configurational and conformational problems solved using cotton effect?

Or

(b) Write a brief account on octant rule.

14. (a) How is the position of $-OH$ group in cholesterol established?

Or

(b) Describe the chemistry of bile acids.

15. (a) Give the mechanism and usefulness of Barton reaction.

Or

(b) (i) How are free radicals generated?

(ii) Write short notes on free radical addition reactions. (3+2)

Part C ($3 \times 10 = 30$)

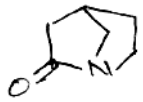
Answer any **three** questions.

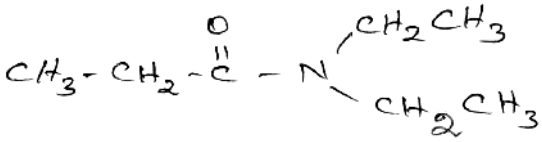
16. (a) Write briefly on :

(i) Structure of the molecular ion.

(ii) IR spectroscopy of organic molecules with respect to inter and intra molecular hydrogen bonding. (4+4)

(b) Which of the following two compounds is expected to have the carbonyl stretching at a higher frequency? Why? (2)

(i) 

(ii) 

17. Write notes on :

(a) Nuclear overhauser effect.

(b) Shift reagents.

(c) Coupling constant. (3.5+3.5+3)

18. Explain how the separation of a mixture of two compounds is affected by column chromatography.

19. (a) Discuss the stereochemistry of steroids.

(b) How is cholesterol converted into testosterone? (6+4)

20. Give an account of synthetic utility of

(a) Phase transfer catalysts.

(b) Merrifield resin.

(c) LDA. (3.5+3.5+3)

AFC–7773

Sub. Code
1MCH3C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Chemistry

INORGANIC CHEMISTRY – III

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define centre of gravity rule.

2. Why are Jahn – Teller distortions automatic for the octahedral geometry of some metal ions?

3. The CO stretching frequency is greater in $[Ni(CO)_4]$ than in $[Fe(CO)4]^{2-}$. Explain.

4. What is Doppler velocity?

5. Predict the ^{19}F–NMR spectrum of BrF_5.

6. Calculate the g-value of cl atoms.

7. Explain the electrical properties of nano particles.

8. Give any two applications of nano tubes.

9. Give an example of a reaction occur at coordinated ligands.

10. Mention the reaction of hydrolysis of aminoacid ester.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Evaluate 10 Dq and β for octahedral d configuration.

Or

(b) What are Tanabe – Sugano diagrams? To what use are they put?

12. (a) Illustrate the use of IT and Raman spectros-copies in distinguishing geometrical and linkage isomers.

Or

(b) What is Mossbauer effect? Explain isomer shift in Mossbauer spectroscopy.

13. (a) Write short notes on :

(i) Contact shift and

(ii) Shift reagents.

Or

(b) Explain the principle of ESR spectroscopy. How are ESR spectrum presented?

14. (a) How are nanomaterials synthesised by sol-gel and coprecipitation methods?

Or

(b) Discuss the structure of carbon nanotubes.

15. (a) Enumerate the mechanism of molecular rearrangement of six coordinated complex.

Or

(b) Describe briefly the mechanism of hydrolysis of amino acid ester.

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. (a) Discuss the electronic spectrum of $[Ni(H_2O)_6]^{2+}$.

    (b) Write short note on change transfer spectra. (5 + 5)

17. (a) Discuss how IR and Raman spectra are used in the study of metal carbonyls and nitrosyls.

    (b) Describe the applications of Mossbauer spectroscopy in the study of iron complexes. (5 + 5)

18. (a) Explain the NMR of paramagnetic metal complexes. (4)

    (b) Write short note on Kramer's degeneracy and zerofield splitting. (6)

19. (a) Discuss in detail about the 1D and 2D nano-structure.

    (b) How are SEM and TEM useful in the characterisation of nano structure? (5 + 5)

20. Explain the mechanism of :

    (a) reactions at coordinated ligands

    (b) hydrolysis of peptides. (5 + 5)

AFC–7774

Sub. Code
1MCH3C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Chemistry

PHYSICAL CHEMISTRY – III

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What do you mean by a block body?
2. Write the quantum mechanical operator for position.
3. Explain :
   (a) eigen value
   (b) eigen function.
4. What are vanishing commutators?
5. Write any two radical function equations.
6. Explain the need for approximation methods.
7. What are thermo setting polymers?
8. Explain the concept of average molecular weight.
9. Why do nano particles exhibit special properties compared to bulk materials?
10. Write the principle of AFM.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Derive the quantum mechanical operator for total energy.

Or

(b) What is compton effect? Derive an expression for the change in wave length of a photon when it is scattered by an electron.

12. (a) Derive schrodinger wave equation.

Or

(b) Show that the operator $\frac{h}{2\pi i}\frac{r}{\partial x}$ is hermitian.

13. (a) Explain zero point energy of simple harmonic oscillator. Write briefly the other allowed energy levels and gaps.

Or

(b) Solve the schrodinger wave equation for an electron moving in a ring.

14. (a) Discuss the mechanism of Zeigler-Natta polymerisation.

Or

(b) Explain the kinetics of free radical polymerisation.

15. (a) Discuss how micro analysis is used in the characterisation of nano materials.

Or

(b) Explain the use of scanning electron microscope (SEM) in the characterisation of nano materials.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Derive the quantum mechanical operator for

    (a) kinetic energy

    (b) potential energy.

17. (a) Solve the schrodinger wave equation for Hydrogen atom.

    (b) State the postulates of quantum mechanics.

18. For a rigid rotator, for a given rotational quantum number J, the degree of degeneracy is (2JH). Justify the statement.

19. Discuss how the molecular weight of a polymer is determined by viscosity method.

20. Discuss the use of following techniques in the characterisation of nanomaterials.

    (a) TEM

    (b) X-ray diffraction.

AFC–7775

Sub. Code
1MCH3E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Chemistry

Elective — POLYMER CHEMISTRY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is meant by poly addition?

2. What is the difference between homo chain and hetero chain polymers?

3. Write any two uses of phenol-formaldehyde.

4. How are urea-formaldehydes prepared?

5. Explain any two intrinsic properties of polymers.

6. What is meant by atactic polymers?

7. Explain about the hydrolytic degradation.

8. Illustrate the mechanical degradation of polymers.

9. Define the term elastomers.

10. Explain film extrusion process of a polymer.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the classification of polymers, with examples.

Or

(b) Explain the mechanism of free radical polymerisation.

12. (a) Explain the preparation and uses of polystyrene and PMMA.

Or

(b) Describe the preparation and uses of polyester and epoxy resin.

13. (a) Explain the stereo chemistry of polymers.

Or

(b) Define glass transition temperature. Explain the factors affecting glass transition temperature.

14. (a) Write short notes on any two polymerisation techniques.

Or

(b) Explain the uses of polymers in electronics and biomedical applications.

15. (a) Distinguish between thermo plastic and thermo setting polymers.

Or

(b) Explain :

(i) Die casting

(ii) Rotational casting.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss the mechanism of polymerisation with respect to :
    (a) Coordination
    (b) Ring opening.

17. Discuss the preparation and uses of :
    (a) Poly ethylene
    (b) Melamine-formaldehyde
    (c) Polytetra fluoroethylene.

18. Determine the molecular weight of polymers :
    (a) Number average method
    (b) Weight average method.

19. Briefly explain any two polymerisation techniques.

20. Write short notes on :
    (a) Reinforcing
    (b) Thermo foaming
    (c) Injection moulding.

AFC–7776

Sub. Code
1MCH4C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fourth Semester

Chemistry

ORGANIC CHEMISTRY — IV

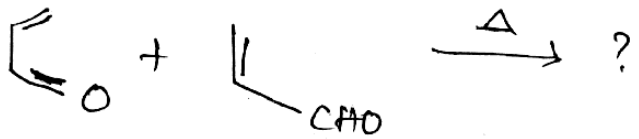
(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** the questions.

1. Draw the structure of RNA.
2. Write the chemistry of oxytocin.
3. State the principle of photo chemistry.
4. Explain electrocyclic reaction.
5. Write the equation of di-pi methane.
6. Give the mechanism of Claisen rearrangements.
7. Write a reaction of imidazole.
8. How will you synthesis of thymine?
9. Formulate the products of the following :



10. Explain activating and blocking group in synthesis.

Section B (5 × 5 = 25)

Answer **all** the questions choosing either (a) or (b).

11. (a) Write a note on chemistry of glutathione.

Or

(b) Describe the classification of proteins and explain the structure of 1°, 2° and 3° protein.

12. (a) Draw and explain the Jablenkshi diagram.

Or

(b) Write short note on photochemistry of Olefins.

13. (a) Discuss the mechanism of vertices rearrangement.

Or

(b) Explain Wagner-Meerwein rearrangement.

14. (a) Describe detailed study of uric acid.

Or

(b) Give the synthesis and reactions of carabazole.

15. (a) Explain planning of synthesis.

Or

(b) Discuss Robinson annelation.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Explain chemistry and structure of DNA.

(b) Write short note on synthesis of merrifield resin.

17. (a) Give the mechanism of Norrish type I and type II reactions.

    (b) Describe sigmatropic rearrangement.

18. Discuss the mechanism of the following reactions.

    (a) Bayer-villegar

    (b) Favorskil rearrangement.

19. Explain the following heterocyclic compound synthesis

    (a) Thiazole

    (b) Cytosine.

20. Describe a schematic analysis of the total synthesis of trans-9-methyl 1-dealone.

AFC–7777

Sub. Code
1MCH4C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fourth Semester

Chemistry

INORGANIC CHEMISTRY – IV

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define the term average line.

2. Explain beta decay of radiations.

3. State the nuclear fission with an example.

4. What is fissile and ferrile isotopes?

5. Give the occurrence of lanthanides.

6. What are the common oxidation state of actinides?

7. Mention any two applications of thermogravimetry.

8. State the theory of electrogravimetric analysis.

9. Write the importance of mainframe computer.

10. What is machine language?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Write a short note on magnetic quantum number.

Or

(b) Explain gamma emission and internal conversion.

12. (a) State and Explain nuclear fission theory.

Or

(b) Discuss the classification of reactors and explain the breeder reactor.

13. (a) Describe the lanthanide contraction and its consequences.

Or

(b) Compare lanthanides and actinides.

14. (a) Write the instrumentation and application of differential thermal analysis.

Or

(b) Describe the cyclic volta metry.

15. (a) Explain the syntax and uses of switch statement with an example.

Or

(b) Write 'C' program to calculate normality, molarity and molality of solution.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Discuss group displacement law.

    (b) Principal and radial quantum number.

17. Describe the applications of radioactive iostopes in various fields.

18. Discuss lanthanides occurrence and extraction from ion exchange method and solvent extraction method.

19. Write the principle, instrumentation and applications of thermograrimetric analysis.

20. Write 'C' programme to determine electronegativity of an atom from bond energy data.

AFC–7778

Sub. Code
1MCH4C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fourth Semester

Chemistry

PHYSICAL CHEMISTRY – IV

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. Explain steady state approximation.

2. State the Marcus theory.

3. What is heterogeneous catalysis?

4. Draw and explain Eadie's plot.

5. Write the Langmuir absorption isotherm equation.

6. Define the term of micro emulsion.

7. State that basic concept of non-equilibrium thermodynamic.

8. Explain catabolism.

9. Write the electronic energy transfer mechanism.

10. What is chemiluminescence?

Section B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Write a note on decomposition of N_2O_5.

Or

(b) Discuss the Hinshelwood theory of unimolecular reaction.

12. (a) Explain Michaelis Menton Kinetics.

Or

(b) Write notes on stopped flow method.

13. (a) Describe Temkin adsorption isotherms.

Or

(b) Compare physisorption and chemisorption.

14. (a) Discuss ATP and its role in bioenergetics.

Or

(b) Describe biological energy conversion of anabolism.

15. (a) Write a note on photosensitization.

Or

(b) Discuss photochemical conversion and storage of solar energy.

Section C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. (a) Evaluate the exploration limits of H_2–O_2 explosive reaction.

    (b) Write a note on kinetics of chain reaction $H_2 - Br_2$.

17. Write a note on :

    (a) Flash photolysis.

    (b) Shock wave method.

18. Derive Gibbs adsorption isotherm.

19. Describe Onsager reciprocal relationship and its application.

20. Derive Stern-Volmer equation and its application.

AFC–7779

Sub. Code
1MCH4C4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fourth Semester

Chemistry

ENVIRONMENTAL CHEMISTRY – IV

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. What is thermal pollution?

2. Define the term of Eco-system.

3. What are causes and effects of global warming?

4. Mention any four air pollutants.

5. Explain water quality index.

6. Mention some important control measure of water pollution.

7. Explain the term of soil waste.

8. What is meant by soil erosion?

9. What are the causes of industrial pollution?

10. Write the name of nuclear power plants in India.

Section B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b)

11. (a) Discuss the major noise pollution and their impact.

Or

(b) Write a brief note on marine pollution.

12. (a) Describe green nouse effect.

Or

(b) Discuss ozone layer depletion.

13. (a) Write a note on ill effects of water pollutants.

Or

(b) Explain sewage and industrial waste treatment.

14. (a) What are the remedial measures should be taken for soil pollution?

Or

(b) Discuss in detail various types of soil pollution.

15. (a) Write a short notes on pulp and paper industry.

Or

(b) Describe nuclear power reactor pollution.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Write a short notes on the following :

(a) Carbon cycle

(b) Hydrological cycle

(c) Nitrogen cycle.

17. (a) Give brief note on acid rain.

    (b) Explain the various methods controlling air pollution.

18. Discuss primary, secondary and tertiray level water treatment.

19. (a) Explain solid waste management.

    (b) Radio active pollution.

20. Explain the following terms :

    (a) Fertilizer and chemical industry

    (b) Agro based industry

    (c) Biochemical effects of lead and mercury.

AFC–7764

Sub. Code
1MCH1C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Chemistry

ORGANIC CHEMISTRY – I

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Between maleic and fumaric acids, which one would you expect to be a stronger acid? Briefly explain.

2. What are carbenes? Give one method for the generation of carbene.

3. What is Bredt's rule?

4. Discuss the stereochemistry of ionic elimination reactions.

5. State Huckel's rule. Explain with an example.

6. What are non-benzenoid aromatic compounds?

7. Draw the Fischer projection formula for (R) – 2 iodo butane and convert it into flying wedge formula.

8. Explain with an example, what is meant by chiral molecule?

9. Write the structure of Vitamin – K_1. How is it synthesised from 2 –methyl –1 , 4 – naphthaquinol.

10. Discuss the stereochemistry of vitamin K_2 .

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Write a brief account on principle of microscopic reversibility.

Or

(b) Discuss the chemistry of nitrenes.

12. (a) (i) Arrange the following compounds in the increasing order of reactivity towards S_N1 displacement.

(1) 2–bromo-2 –methylbutane.

(2) 1– bromopentane

(3) 2–bromo–2 phenylpentane

(4) 1–bromo–1– methylcyclohexane.

(ii) Explain $S_N i$ reaction with a suitable example. (3+2)

Or

(b) (i) Under what conditions an E2 mechanism change over to an $E1cB$ mechanism. Explain with suitable examples.

(ii) What are ambident substrates? Give two examples. (3+2)

13. (a) Discuss the aromaticity of annulenes.

Or

(b) Explain the chemistry of cyclopentadienyl anion.

14. (a) State and explain Prelog's rule with two examples.

Or

(b) (i) What do you understand by gauche and anti conformations?

(ii) Define the term optical activity. (3+2)

15. (a) Predict the products

(i) Riboflavin $\xrightarrow[h\gamma]{NaOH} A \xrightarrow[HCl]{Ba(OH)_2} B$

(ii) Pyriodoxine

$$\xrightarrow{CH_2N_2} A \xrightarrow{Ba(MnO_4)_2} B \xrightarrow[resorcinol]{Fused} C.$$ (2+3)

Or

(b) Give one method of synthesis of pyridoxin.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) What are carbocations? How do you account for the relative stability of primary, secondary and tertiary carbocations?

(b) Explain the concept of hard and soft acids and bases with examples. (5+5)

17. (a) Narrate the effect of solvent on S_N1 and S_N2 reactions.

(b) Write a note on nucleophilic substitution at a vinylic carbon. (6+4)

18. (a) Explain alternant and non–alternant hydrocarbons.

(b) Describe the homo and antiaromaticity in 2 and 4π electron systems. (5+5)

19. (a) Write a note on the geometrical isomerism of the oximes and the method used for the determining their configuration.

(b) Explain with examples the superiority of R–S system of configuration over D-L system. (5+5)

20. Discuss the degradative evidences that establish the structure of vitamin B_1 confirm its structure by synthesis.